# 2016年真题参考答案

## 一、选择题

(1)B. (2)D. (3)B. (4)A. (5)C. (6)C. (7)A. (8)C.

## 二、填空题

(9)6. (10)$\sin 1-\cos 1$. (11)$-\mathrm{d}x+2\mathrm{d}y$. (12)$\frac{1}{3}-\frac{2}{3\mathrm{e}}$.

(13)$\lambda^4+\lambda^3+2\lambda^2+3\lambda+4$. (14)$\frac{2}{9}$.

## 三、解答题

(15)$\mathrm{e}^{\frac{1}{3}}$.

(16)(Ⅰ)$Q(p)=1\,200-10p$.

(Ⅱ)当$p=100$万元时,边际收益为80万元. 其经济意义为:当$p=100$万元,$Q=200$件时,销售第201件商品所得的收益为80万元.

(17)$f'(x)=\begin{cases}4x^2-2x, & 0<x<1,\\ 2x, & x\geqslant 1.\end{cases}$最小值$f\left(\frac{1}{2}\right)=\frac{1}{4}$.

(18)$f(x)=-\frac{\mathrm{e}^{-x}+\mathrm{e}^{x}}{2}$.

(19)和函数$s(x)=\begin{cases}x\ln\left(\frac{1+x}{1-x}\right)+\ln(1-x^2), & x\in(-1,1),\\ 2\ln 2, & x=\pm 1.\end{cases}$收敛域为$[-1,1]$.

(20)(Ⅰ)$a=0$. (Ⅱ)通解为$k(0,1,-1)^{\mathrm{T}}+(1,-2,0)^{\mathrm{T}}$,其中$k$为任意常数.

(21)(Ⅰ)$\boldsymbol{A}^{99}=\begin{pmatrix}2^{99}-2 & 1-2^{99} & 2-2^{98}\\ 2^{100}-2 & 1-2^{100} & 2-2^{99}\\ 0 & 0 & 0\end{pmatrix}$.

(Ⅱ)$\boldsymbol{\beta}_1=(2^{99}-2)\boldsymbol{\alpha}_1+(2^{100}-2)\boldsymbol{\alpha}_2$,$\boldsymbol{\beta}_2=(1-2^{99})\boldsymbol{\alpha}_1+(1-2^{100})\boldsymbol{\alpha}_2$,$\boldsymbol{\beta}_3=(2-2^{98})\boldsymbol{\alpha}_1+(2-2^{99})\boldsymbol{\alpha}_2$.

(22)(Ⅰ)$f(x,y)=\begin{cases}3, & 0<x<1,x^2<y<\sqrt{x},\\ 0, & \text{其他}.\end{cases}$

(Ⅱ)$U$与$X$不相互独立.理由见解析.

(Ⅲ)$F(z)=\begin{cases}0, & z\leqslant 0,\\ \frac{3}{2}z^2-z^3, & 0<z\leqslant 1,\\ 2(z-1)^{\frac{3}{2}}-\frac{3}{2}z^2+3z-1, & 1<z\leqslant 2,\\ 1, & z>2.\end{cases}$

(23)(Ⅰ)$f_T(t)=\begin{cases}\frac{9t^8}{\theta^9}, & 0<t<\theta,\\ 0, & \text{其他}.\end{cases}$ (Ⅱ)当$a=\frac{10}{9}$时,$E(aT)=\theta$.